

CS & IT ENGINEERING



Computer Network

Routing

Lecture No. - 05

By - Abhishek Sir





Recap of Previous Lecture



Topic

Distance Vector Routing





Topics to be Covered



Topic

Distance Vector Routing

Topic

Routing Protocols

ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
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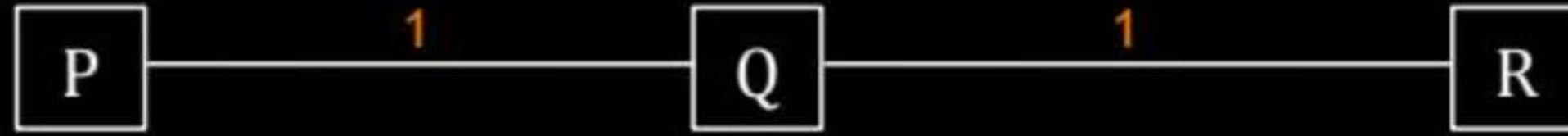
#Q. Consider a network with three routers P, Q, R shown in the figure below. All the links have cost of unity.



The routers exchange distance vector routing information and have converged on the routing tables, after which the link Q-R fails. Assume that P and Q send out routing updates at random times, each at the same average rate. The probability of a routing loop formation (rounded off to one decimal place) between P and Q, leading to count-to-infinity problem, is _____.

[GATE-2022, NAT, 1-Mark]

$$\text{Ans} = \frac{1}{2} = 0.5$$



Node	Cost	Via
P	0	P
Q	1	Q
R	2	Q

Node	Cost	Via
P	1	P
Q	0	Q
R	1	R

Node	Cost	Via
P	2	Q
Q	1	Q
R	0	R

CASE I :- No any Routing Loop formation



Node	Cost	Via
P	0	P
Q	1	Q
R	2	Q

Node	Cost	Via
P	1	P
Q	0	Q
R	∞	

Node	Cost	Via
P	∞	
Q	∞	
R	0	R



P	1
Q	0
R	∞

CASE II :-

Routing Loop Exist.



Node	Cost	Via
P	0	P
Q	1	Q
R	2	Q

Node	Cost	Via
P	1	P
Q	0	Q
R	∞	

Node	Cost	Via
P	∞	
Q	∞	
R	0	R

→

P	0
Q	1
R	2

←

P	1
Q	0
R	∞



Topic : Distance Vector Routing



→ **Problem** : May suffer with “Count to infinity” problem
[Link to infinity problem, occur due to routing loop in DVR]

Solution 1 : Advertise routing path in ‘Distance Vector’
[Send ‘Next Hop’ (via) with distance vector]

Solution 2 : “Split horizon hack”
[Do not advertise the distance to a neighbor,
if the neighbor is the Next Hop for corresponding entry]

Solution 3 : “Poison reverse”
[Advertise infinite distance to a neighbor,
if the neighbor is the Next Hop for corresponding entry]

#Q. Two popular routing algorithms are Distance Vector (DV) and Link State (LS) routing. Which of the following are true?

(S1) Count to infinity is a problem only with DV and not LS routing TRUE

(S2) In LS, the shortest path algorithm is run only at one node FALSE

(S3) In DV, the shortest path algorithm is run only at one node FALSE

(S4) DV requires lesser number of network messages than LS TRUE

[GATE-2008]

(A) S1, S2 and S4 only

(B) S1, S3 and S4 only

(C) S2 and S3 only

✓ (D) S1 and S4 only

Max^m No. of messages in DV (per iteration)
= $2 \times \text{No. of edges (links)}$

Ans: D

#Q. Consider the following three statements about link state and distance vector routing protocols, for a large network with 500 network nodes and 4000 links.

[S1] The computational overhead in link state protocols is higher than in distance vector protocols. TRUE

[S2] A distance vector protocol (with split horizon) avoids persistent routing loops, but not a link state protocol. FALSE

[S3] After a topology change, a link state protocol will converge faster than a distance vector protocol. TRUE

Which one of the following is correct about S1, S2, and S3 ?

[GATE-2014, Set-1, 1-Mark]

(A) S1, S2, and S3 are all true

(B) S1, S2, and S3 are all false

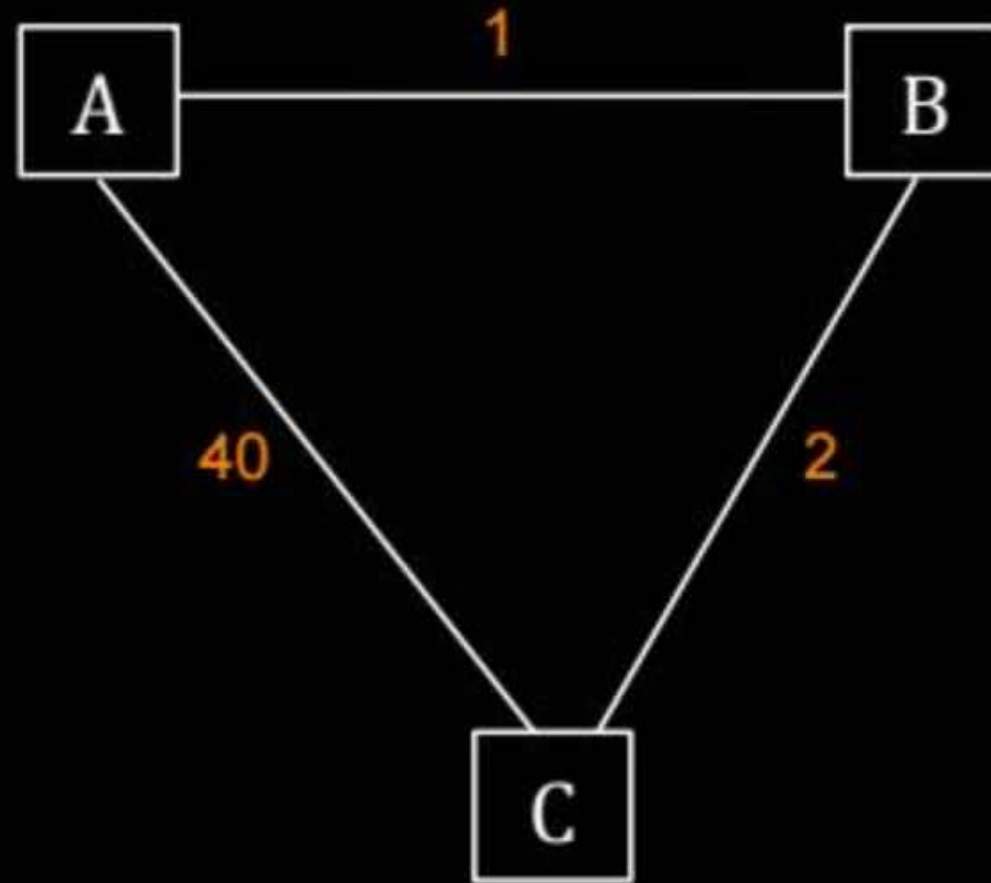
(C) S1 and S2 are true but S3 is false

☒ (D) S1 and S3 are true, but S2 is false

Ans: D

DV Routing (Without split horizon hack) :- [Stabilized Distance Vector]

Node	Cost	Via
A	0	A
B	1	B
C	3	B

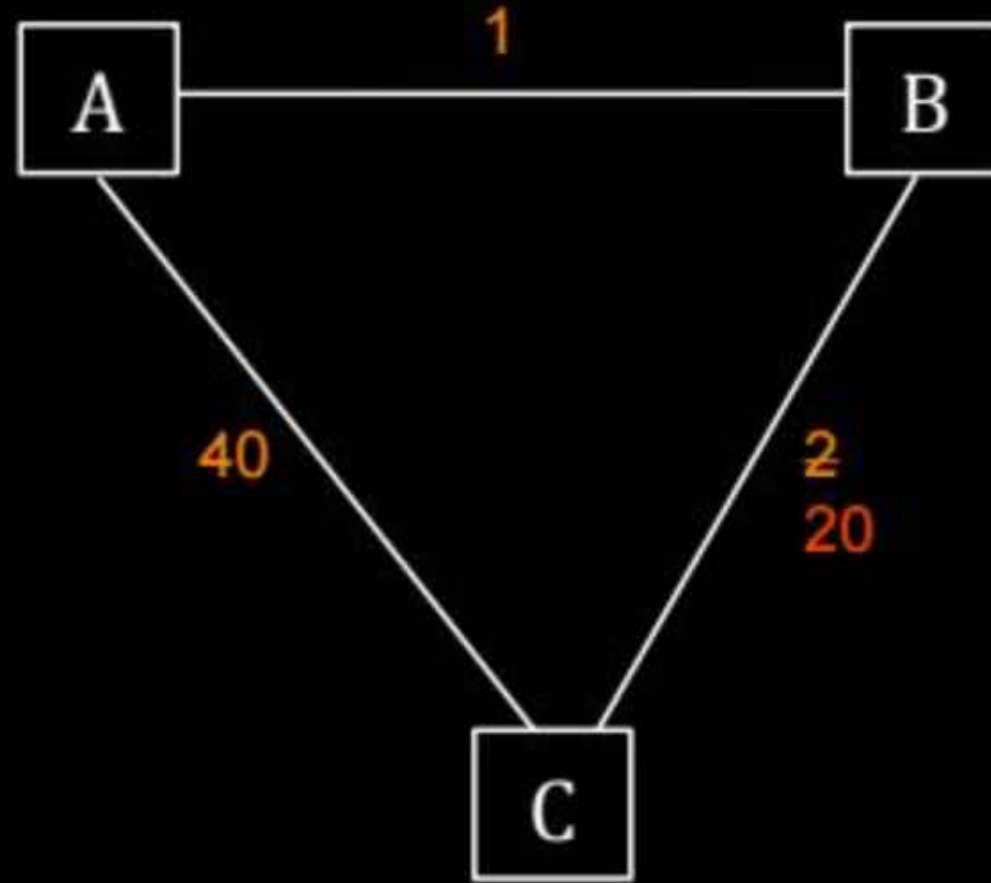


Node	Cost	Via
A	1	A
B	0	B
C	2	C

Node	Cost	Via
A	3	B
B	2	B
C	0	C

DV Routing (Without split horizon hack) :- [Link Cost Increases]

Node	Cost	Via
A	0	A
B	1	B
C	3	B



Node	Cost	Via
A	1	A
B	0	B
C	20	C

Node	Cost	Via
A	3	<u>B</u>
B	20	B
C	0	C

	Router A	Router B
[Link Cost Increases]	$C = 3$, [Via B]	$C = 20$, [Via C]
→ (First) iteration	$C = 21$, [Via B]	$C = 4$, [Via A]
Second iteration	$C = 5$, [Via B]	$C = 20$, [Via C]
Third iteration	$C = 21$, [Via B]	$C = 6$, [Via A]
⋮		
16 th iteration	$C = 19$, [Via B]	$C = 20$, [Via C]
17 th iteration	$C = 21$, [Via B]	$C = 20$, [Via C]

18th iteration $C = 21$, via B $C = 20$, via C



Topic : Distance Vector Routing

- Each router wait for
either “changes in adjacent link cost”
or “receives new “Distance Vector” estimate from any neighbor”
- It update its own “Distance Vector” estimate using Bellman-Ford equation
- if any update in its own “Distance Vector” estimate
then send own updated “Distance Vector” estimate to their neighbors
else “do nothing”

Asynchronous

Statement for Linked Answer Questions.

Consider a network with five nodes, N1 to N5, as shown below. The network uses a Distance Vector Routing protocol. Once the routes have stabilized, the distance vectors at different nodes are as following.

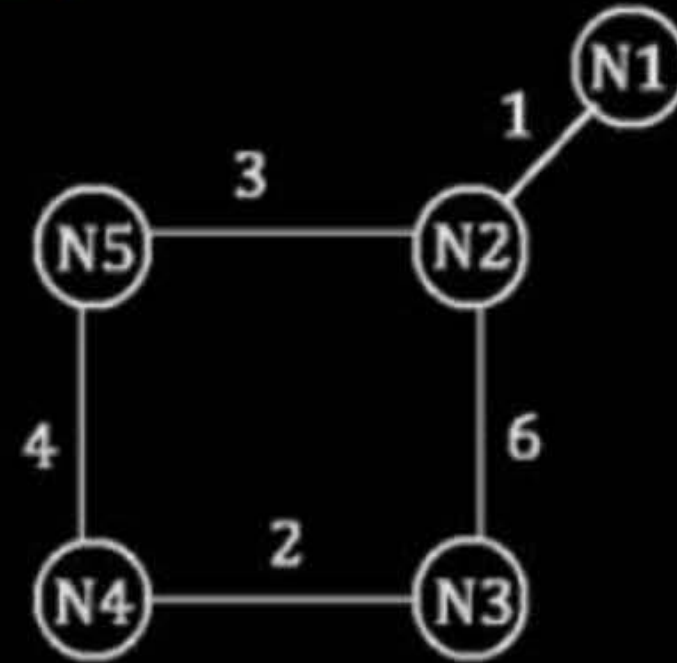
N1 : (0, 1, 7, 8, 4)

N2 : (1, 0, 6, 7, 3)

N3 : (7, 6, 0, 2, 6)

N4 : (8, 7, 2, 0, 4)

N5 : (4, 3, 6, 4, 0)



Each distance vector is the distance of the best known path at that instance to nodes, N1 to N5, where the distance to itself is 0. Also, all links are symmetric and the cost is identical in both directions. In each round, all nodes exchange their distance vectors with their respective neighbors. Then all nodes update their distance vectors. In between two rounds, any change in cost of a link will cause the two incident nodes to change only that entry in their distance vectors.

#Q. The cost of link N2-N3 reduces to 2 (in both directions). After the next round of updates, what will be the new distance vector at node, N3?

[GATE-2011]

$N_3(DV)$

Router	N_1	N_2	N_3	N_4	N_5	
Distance	3	2	0	2	5	
Via	N_2	N_2	N_3	N_4	N_2	

(A) (3, 2, 0, 2, 5)

(B) (3, 2, 0, 2, 6)

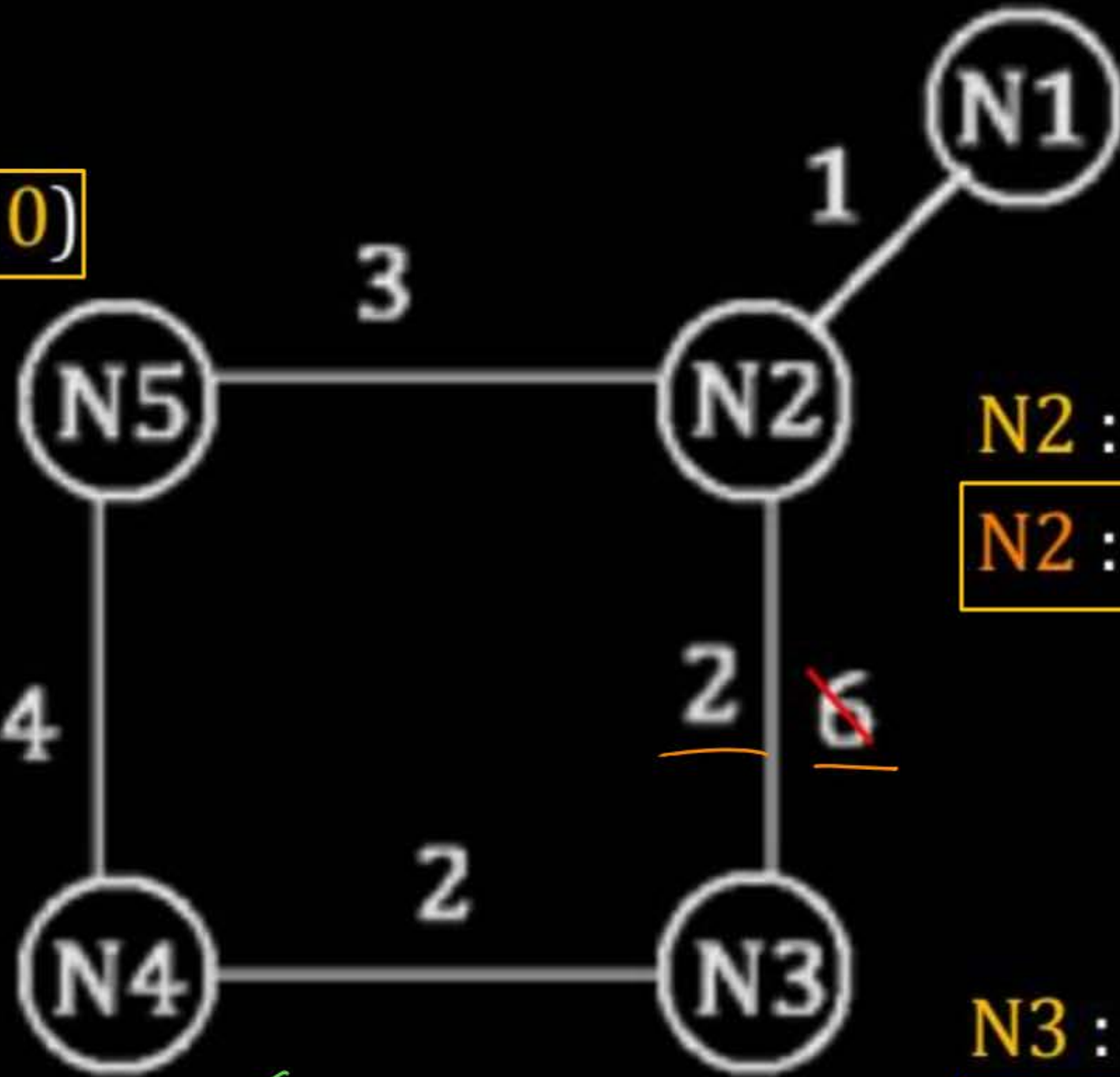
(C) (7, 2, 0, 2, 5)

(D) (7, 2, 0, 2, 6)

Ans: A

N5 : (4, 3, 6, 4, 0)

N1 : (0, 1, 7, 8, 4)



N2 : (1, 0, 6, 7, 3)

N2 : (1, 0, 2, 7, 3) ✓

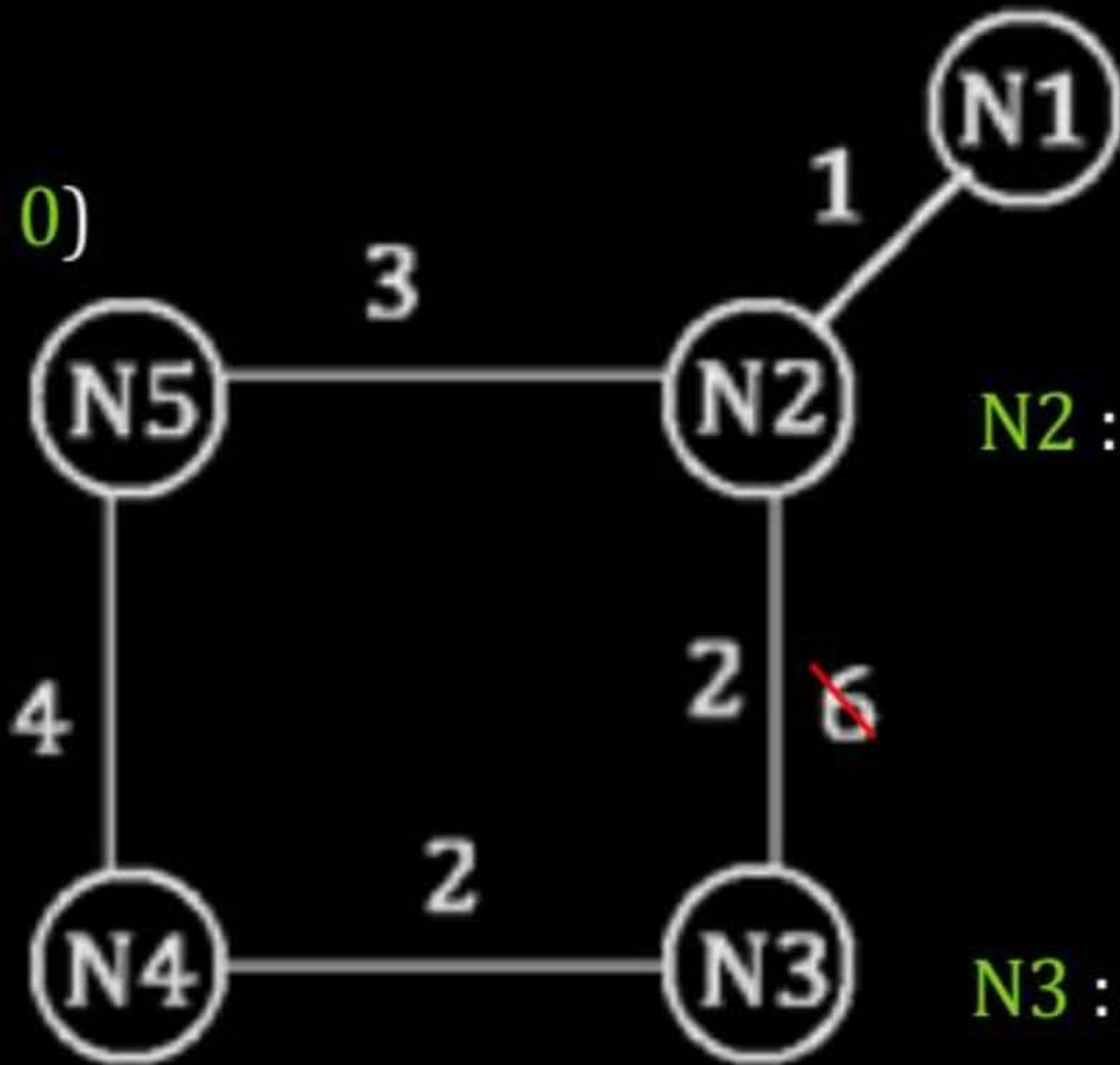
N3 : (7, 6, 0, 2, 6)

N3 : (7, 2, 0, 2, 6)

N4 : (8, 7, 2, 0, 4) ✓

$N1 : (0, 1, 3, 8, 4)$

$N5 : (4, 3, 5, 4, 0)$



$N2 : (1, 0, 2, 4, 3)$

$N3 : (3, 2, 0, 2, 5)$

$N4 : (8, 4, 2, 0, 4)$



Topic : OSPF



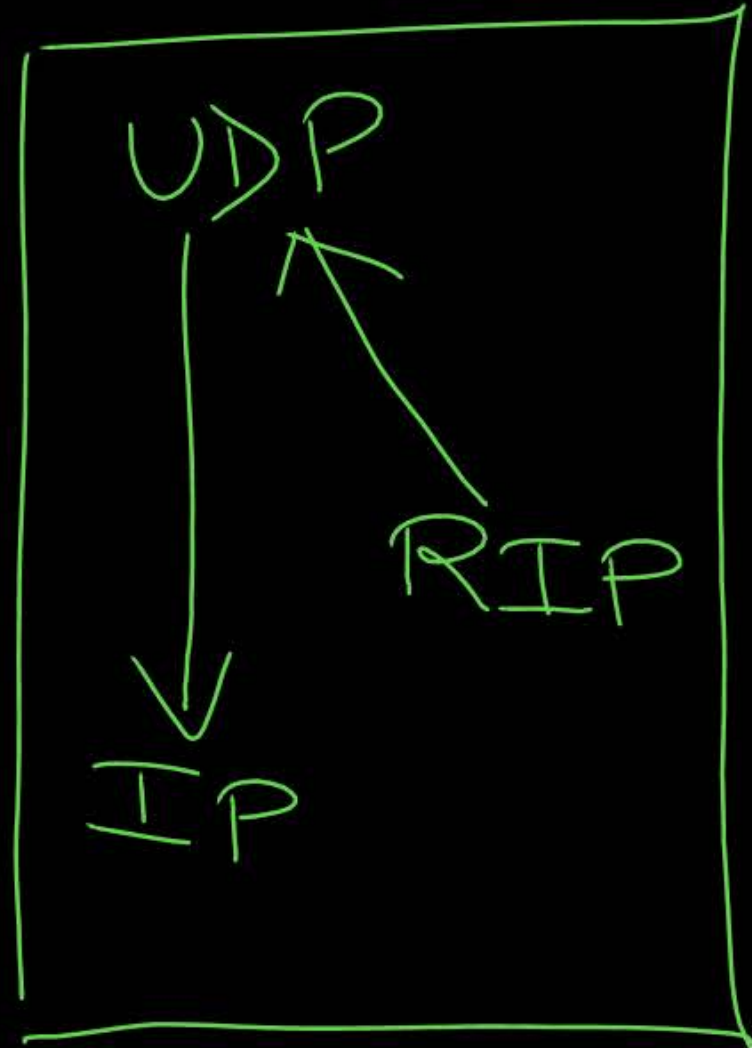
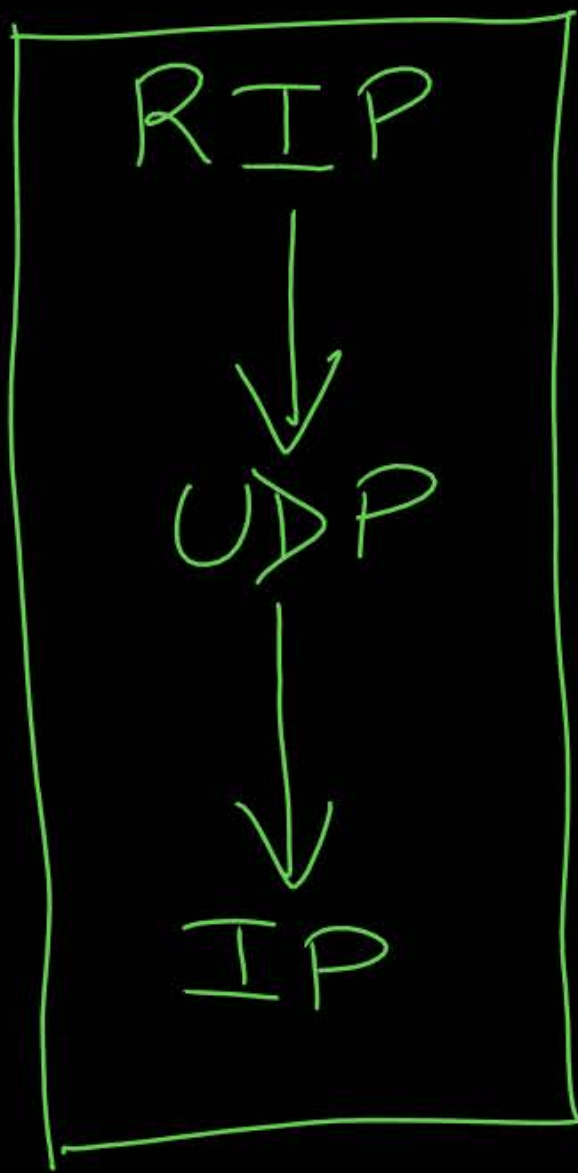
- Routing Protocol operate at Network Layer
- Classic Link State Routing
- OSPF packets are directly encapsulated into IP datagram payload field
[OSPF → IP]
- Link cost metric : Bandwidth or delay ✓
- Each router has full topology
[Uses Dijkstra's algorithm to compute 'forwarding table']



Topic : RIP



- RIP : Routing Information Protocol
- Routing Protocol operate at Network / Application Layer
- Classic Distance Vector Routing
[Distance Vector exchanged every 30 second] $\Rightarrow$ Synchronous DV Routing
- Based on the Bellman-Ford algorithm
- RIP packets are directly encapsulated into UDP datagram payload field
[RIP $\rightarrow$ UDP $\rightarrow$ IP]
- Link cost metric : Hop count



#Q. Consider the following statements about the routing protocols, Routing Information Protocol (RIP) and Open Shortest Path First (OSPF) in an IPv4 network.

- I. RIP uses distance vector routing (RIP-DV) TRUE
- II. RIP packets are sent using UDP (RIP → UDP → IP) TRUE
- III. OSPF packets are sent using TCP FALSE (OSPF → IP)
- IV. OSPF operation is based on link-state routing TRUE (OSPF-LS)

Which of the following above are CORRECT?

[GATE-2017, Set-2, 1-Mark]

(A) I and IV only

(B) I, II and III only

✓ (C) I, II and IV only

(D) II, III and IV only

Ans: C

(IIT-R)



Topic : BGP



- BGP : Border Gateway Protocol
- EGP : Exterior Gateway Protocol
[Inter-ISP routing]
- Path Vector Routing
[Routing messages contains complete routes]
- Routing Protocol operate at Application Layer / Network
- BGP packets are directly encapsulated into TCP segment payload field
[BGP → TCP → IP]



2 mins Summary



Topic

Distance Vector Routing ✓

Topic

Routing Protocols ✓

Source Routing
Hierarchical Routing
[Traffic Shaping]



THANK - YOU

